SUR 2025/2026 Project

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1. Augmentation

The training pool is small (30 base paired stems of *m431* across two sessions, ~190 non-target stems from disjoint speakers) and the assignment warns the eval set will contain input-quality variations not seen at training. Offline augmentation (`scripts/augment_dataset.py`) addresses both: it enlarges the effective training set and regularises against acoustic and visual distortions. Audio augmentation applies time stretch (0.9–1.1×), gain (±6 dB), polarity flip ($p=0.5$), peaking-band equalisation (EQ; ±3 dB at 200–4000 Hz) and additive White Gaussian Noise (WGN) (SNR 15–30 dB). image augmentation applies horizontal flip ($p=0.5$), rotation (±10°), translation (±5 %), brightness and contrast scaling (0.85–1.15×) and per-pixel WGN ($\sigma \in [0,0.04]$). Each base target sample receives 15 augmented variants and each non-target receives 5.

2. Evaluation Metrics

Four quantities are reported throughout. **AUC** is the primary rank statistic, threshold-free and robust to class imbalance. Most variant-selection decisions (audio feature subsets, image regularisation, GMM-UBM mixture counts) were made on AUC. **Equal Error Rate (EER) and the EER threshold** summarise the symmetric-cost operating point under prior 0.5. Drift of the EER threshold away from zero flags miscalibration, and we rejected the MAP-adapted GMM-UBM variant on this signal alone (its EER threshold sat at $+0.25$ across the entire τ sweep). **Confusion at *logit = 0*** (TP/FP/FN/TN) gives the operationally relevant quantities under the assignment's hard rule. **Precision and F_1** become decisive when stress-dev separates the fusion variants: `fusion_3way` reaches precision 0.70 and $F_1 = 0.70$ on stress dev versus 0.31 / 0.47 for the more recall-favouring fusions, which is what motivates keeping it in the submitted lineup despite a slightly lower stress AUC than `fusion_hybrid_image`.

3. Image System

Image features are 9-bin Histograms of Oriented Gradients over 8×8-pixel cells with 2×2-cell L2-Hys normalised blocks on 80×80 grayscale crops, producing 2916-d vectors. A balanced logistic regression with L2 regularisation ($C=0.1$) classifies them. Hard decisions threshold *decision_function* at zero. The progression from the syllabus baseline isolates three design changes:

Step	Description	AUC	EER
1	Baseline: pixels → PCA(D=21) → Gaussian LDA + 2 IRLS	0.943	20.00 %
2	HOG → PCA(D=21) → Gaussian LDA + 2 IRLS	0.977	9.17 %
3	HOG (full 2916-d) → balanced LogReg, $C=0.1$, no augmentation	0.903	21.67 %
4	+ offline augmentation (submitted)	1.000	0.00 %

Step 1→2 isolates the HOG vs. raw-pixel feature gain at fixed dimensionality. Step 2→3 is a *regression*: dropping PCA in favour of full-dimensional HOG with L2 leaves the threshold-zero hard rule mis-calibrated at only 10 target training images. Step 3→4 shows that augmentation is what makes the high-dimensional architecture viable; full-dimensional + L2 in isolation is worse than the PCA-bridged baseline classifier of step 2.

4. Audio System

Both audio systems share the same frontend: an energy + zero-crossing-rate voice activity detector (VAD), 13 mel-frequency cepstral coefficients at 25 ms / 10 ms framing with per-sample cepstral mean–variance normalisation, and probabilistic-YIN F_0 estimation over voiced frames in 60–400 Hz. From the F_0 track and frame energies we compute five prosodic statistics (*log F_0* mean and standard deviation, jitter, shimmer, voicing ratio).

Path A, aggregated features and logistic regression. The 5 prosodic statistics and the 13 MFCC means form an 18-d sample-level vector classified by balanced LogReg (*C=1.0*). This is shipped as `audio_logreg.txt`.

Path B, Gaussian Mixture Model with universal background, MAP-free. Following we fit a 16-mixture diagonal-covariance universal background model (UBM) on non-target frames and a 4-mixture diagonal-covariance target GMM on *m431* frames, *independently* by EM. We chose MAP-free over MAP adaptation because ~32 k augmented target frames are plentiful at the per-component level ($\alpha_k \to 1$ regardless of relevance factor), and a relevance-factor sweep over $\tau \in \{5, 10, 20, 100, 500, 2000, 10000\}$ produced an AUC plateau of 0.98 with the threshold-zero hard rule miscalibrated (EER threshold drifted to *+0.25*). The per-sample score is the frame-averaged log-likelihood ratio. We did not attempt i-vector front-end factor analysis, the standard upgrade over GMM-UBM, because it requires hundreds of background speakers for the total-variability matrix to be estimable, whereas our pool has 19 distinct non-target speakers.

The VAD-empty pathology. Auditing eval revealed 79 of 1346 samples (5.9 %) producing identical audio score ≈ *+1.535*. These extract to all-zero feature vectors: under heavy gain attenuation or short voiced segments, the energy-relative VAD trims every frame, leaving prosodic and MFCC statistics at their zero-fallback. The trained classifier maps this zero input to a bias-driven default that happened to be positive, mislabelling all 79 as target. Spectral-flatness and voiced-fraction analysis confirmed all 79 are real speech, not white noise (median voiced fraction 0.60); the sentinel default was discarding genuine information. The same pathology contaminates training: 121 of 1112 augmented training samples also extract to zero vectors, with the target class disproportionately affected (81 zero-frame target augmentations vs. 40 non-target), so the model also *learned* a small "zero ⇒ target" correlation as a class-asymmetric augmentation artefact.

Hybrid VAD-on / VAD-off fallback. The fix is a routing scheme (`scripts/audio_gmm_hybrid.py`): an auxiliary 16+4 GMM-UBM is trained on the same MFCC frames *without* applying VAD; at scoring time the submitted VAD-on model is consulted first, and the no-VAD model takes over only when VAD has trimmed every frame. The hybrid is shipped as `audio_gmm_hybrid.txt` and short-circuits both the eval-time sentinel defaults and the train-time spurious correlation at scoring.

The audio progression covers both clean and stress development splits:

Step	Description	Clean AUC	Stress AUC
1	Baseline: course MFCC + 2 GMMs M=30 (no VAD, no CMVN)	0.978	0.799
2	+ frame-energy VAD + per-sample CMVN, same 2 GMMs M=30	1.000	0.730
3	Submitted GMM-UBM (MAP-free, sentinel for VAD-empty)	0.997	0.463
4	Hybrid VAD-on/off (submitted)	0.997	**0.870**

VAD and CMVN saturate clean dev (steps $1 \to 2$) but degrade the stress split because they introduce the failure mode that step 4 must repair. The hybrid recovers the baseline's stress robustness without sacrificing clean-dev performance.

5. Multimodal Fusion

Fusion is performed at the score level. For each per-modality system, dev-set scores are z-normalised; a balanced logistic regression is fitted on the concatenated z-scores and the ground-truth labels; the classifier is then applied to z-normalised eval scores. Score-level was preferred to feature-level because the audio (18-d) and image (2916-d) spaces are extremely imbalanced: stacking them with ~1100 training samples invites the curse of dimensionality, whereas a 2- or 3-feature fusion logreg has too few parameters to overfit the 70-sample dev split. We submit three variants: `fusion_logreg.txt` over $(z(\text{audio_logreg}), z(\text{image}))$; `fusion_hybrid_image.txt` over *(z(audio_gmm_hybrid), z(image))*; and `fusion_3way.txt` over $(z(\text{audio_logreg}), z(\text{audio_gmm_hybrid}), z(\text{image}))$. The three-way variant tests whether the two audio paths capture complementary signal, since aggregated 18-d features and frame-level GMM likelihoods are structurally different objects.

6. Validation Strategy

Clean `*_dev` (n=70) saturates for image and fusion, so variant selection ran on a *stress* dev built by re-augmenting dev with a *disjoint* envelope (gain *-15* to *-8* dB, SNR 3–13 dB, time-stretch 0.78–0.88, image brightness 0.45–0.70, per-pixel noise $\sigma \in [0,0.10]$), which converts each sample from interpolation to extrapolation under the trained model. Each variant in the per-modality sweeps was promoted into the submitted lineup on its stress AUC, with stress EER and threshold-zero confusion as tiebreakers. The 50/50-prior decision rule is enforced at training (`class_weight='balanced'`) and inference (`hard_threshold_logit = 0.0`).

7. Results

The submission ships one image system, two audio systems and three fusion variants, 6 result files in total.

Clean dev (n = 70: 10 target / 60 non-target):

System	AUC	EER	TP	FP	FN	TN
image_logreg	1.000	0.00 %	10	1	0	59
audio_logreg	0.983	3.33 %	7	0	3	60
audio_gmm_hybrid	0.997	1.67 %	9	0	1	60
fusion_logreg	1.000	0.00 %	10	0	0	60
fusion_hybrid_image	1.000	0.00 %	10	0	0	60
fusion_3way	1.000	0.00 %	10	0	0	60

Stress dev (n = 170: 50 target / 120 non-target):

System	AUC	EER	TP	FP	FN	TN
image_logreg	0.607	40.00 %	35	60	15	60
audio_logreg	0.521	47.08 %	50	113	0	7
audio_gmm_hybrid	0.870	24.08 %	47	69	3	51
fusion_logreg	0.625	39.17 %	50	111	0	9
fusion_hybrid_image	0.875	20.42 %	50	107	0	13
fusion_3way	0.860	21.83 %	35	15	15	105

On the unlabelled eval set (n = 1346), predicted-target counts under the threshold-zero rule are 127 (image_logreg), 153 (audio_logreg), 139 (audio_gmm_hybrid), 142 (fusion_logreg), 178 (fusion_hybrid_image) and 139 (fusion_3way). DET curves are saved at `artifacts/curves*.png`; we prefer DET over ROC because its log-deviate axes spread the high-performance region that AUC saturation collapses into the corner of the ROC.

All three fusion variants saturate clean dev. Stress dev separates them clearly: `fusion_hybrid_image` maximises AUC (0.875); `fusion_3way` is the only system whose stress confusion matrix is balanced (precision 0.70 versus 0.31 for the alternatives), trading a small AUC decrement for substantial precision gain. Under an operating point that penalises false positives more strongly than the assignment's symmetric costs, `fusion_3way` would be the strongest single submission.

8. Discussion

The most surprising finding is the *clean–stress inversion* in the audio progression: the simple syllabus baseline (no VAD, no CMVN) handles the stress split substantially better than the submitted preprocessed pipeline does. The mechanism is that the submitted VAD trims every frame on heavily attenuated audio (161 of 170 stress samples and 79 of 1346 eval samples produce all-zero feature vectors); without VAD, the attenuated signal is still scored on its remaining content. The hybrid fallback recovers the baseline's stress robustness on the GMM-UBM path while preserving clean-dev saturation. We did not attempt to remove or retune the VAD on the LogReg path because disabling it costs ~35 AUC points on clean dev, and a routing fix restricted to the small known-degenerate slice is a smaller intervention than a full pipeline retune. A from-scratch convolutional network was ruled out for reasons analogous to i-

vectors: a small CNN (~50 k parameters) needs an order of magnitude more target examples than the 320 augmented samples we have.

9. Reproduction

```
uv run scripts/augment_dataset.py --config configs/augment_audio.toml
uv run scripts/augment_dataset.py --config configs/augment_image.toml
uv run scripts/train.py --config configs/audio_baseline.toml
uv run scripts/train.py --config configs/image_baseline.toml
uv run scripts/train.py --config configs/fusion_baseline.toml
uv run scripts/evaluate.py --config configs/audio_baseline.toml --split eval
uv run scripts/evaluate.py --config configs/image_baseline.toml --split eval
uv run scripts/evaluate.py --config configs/fusion_baseline.toml --split eval
uv run scripts/audio_gmm_pipeline.py --mode map_free
uv run scripts/audio_gmm_hybrid.py
uv run scripts/fix_vad_empty.py
uv run scripts/fusion_hybrid_image.py
uv run scripts/fusion_3way.py
```

All randomness is seeded; reproduction is deterministic from a fixed seed.